Split plan: moving the prognostic material and the F-family out of paper 02

2026-10-29 19:16 PDT

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Status. Plan only. No manuscript, driver or data file has been changed. Line numbers refer to `report.Rmd` at commit 8e810ca9d (2026-10-29). They will drift as soon as the first cut is made, so work from the section labels.

Notation

Symbols follow `analysis/report/NOTATION.md`. Code names are in `monospace`. Two symbols have no NOTATION.md entry yet. They are marked *proposed* and should be added when the split is adopted. Paper 02's current text uses retired forms (c_{pb}, c_{tv}, c_{bl}, D_e). The cut should rename them to the forms below. Do not declare them as departures.

Symbol	Meaning	Code
B_i, B_i^c	biomarker of participant i; centered at the trial's baseline sample mean	`bm`, `bmc`
w_t	calendar week of visit t	`t`
c_{bm}	biomarker-response correlation on drug: the interaction under covariance moderation	`c_bm`, `c.bm`
c_{bm}^*	ceiling: the largest c_{bm} for which every path's correlation matrix is positive definite	
$c_{bm,PB}$, $c_{bm,TV}$, $c_{bm,BL}$	prognostic correlations: of the biomarker with the placebo-response component, the natural-history component, and the baseline level	`c_pb`, `c_tv`, `c_bl`; `c.bm.pb`
b_{TV} (*proposed*)	constant natural-history mean shift per SD of the biomarker; absorbed by the biomarker main effect	`b_tv`, `b.bm.tv`

Symbol	Meaning	Code
b^{tr}_{TV} (*proposed*)	trend-pathway coefficient: shift of $b^{\text{tr}}_{TV}\,\sigma_{TV}\,z_i\,g(w_t)$, with z_i the standardized biomarker and g the TV mean course scaled to peak 1	`b_tv_trend`, `b.bm.tv.trend`
c_1	cross-component correlation at the same visit (refined process 0.2)	`c.cf1t`, `c.cfct`
$t_{1/2}$, $t^a_{1/2}$	carryover half-life of the DGP; half-life assumed by the analysis	`t1half`; `h`, `hl_analysis`
$D_{bc,it}$	exposure-weighted drug indicator built at $t^a_{1/2}$	`Dbc`
η_t	design expectancy: 1 open label, 0.5 blinded, 0 at baseline	`e`; regressor `De`
β_η, $\beta_{bm:\eta}$	coefficients of the expectancy terms η_t and $B^c_i\eta_t$	`De`, `bmc:De`
κ	mean model-based SE divided by the empirical SD of the estimates	`kappa`
$\hat{\pi}$, MCSE	estimated rejection rate; its Monte Carlo SE, $\sqrt{\hat{\pi}(1-\hat{\pi})/n_{\text{reps}}}$	`reject`, `mcse`

Paper 02's `bm` is already centered by the DGP (mean 0), so `bm` and `bmc` coincide in its drivers. The remedy term is written `bmc:t` ($B^c_i w_t$) throughout this plan, and its quadratic variant adds $B^c_i w_t^2$.

Glossary

- **Covariance moderation.** The data-generating convention in which the interaction is a correlation between the biomarker and the drug response, present on drug and decaying off drug. Paper 02 calls it the "correlation encoding"; code labels it `mvn`.
- **Refined process.** The recommended DGP: graded coupling, separable AR(1) with $\rho = 0.7$ and $c_1 = 0.2$, anchored carryover, and a stop on

non-positive-definite matrices. Paper 02 calls it the "extended DGP". **Published process:** Hendrickson's code at commit 58b32a9.

- **Predictive biomarker.** A biomarker that changes the drug response ($c_{bm} \neq 0$). **Prognostic biomarker:** one associated with the outcome through a channel other than the drug response. A biomarker can be both.
- **Prognostic pathway.** One channel of prognostic association:
  - **prog-BL:** $c_{bm,BL}$, baseline level.
  - **prog-TV-const:** $c_{bm,TV}$, natural-history component.
  - **prog-PB:** $c_{bm,PB}$, placebo-response component.
  - **constant shift:** b_{TV}, stored as `prog-TV-grow` despite being constant.
  - **trend pathway:** b_{TV}^{tr}, a natural-history mean shift that grows over the trial.

  The first three are correlations and are bounded by feasibility. The last two are mean shifts and are not.
- **Feasibility bound.** The largest prognostic correlation, given c_{bm}, for which the joint correlation matrix is positive definite on every path. It has a closed form via the Schur complement (`docs/65`).
- **Silent repair; repair artifact.** Before 2026-10-26 the drivers replaced a non-positive-definite matrix with a nearby positive-definite one without warning. In the strength-0.6 prognostic cells this attenuated the interaction and showed up as a power loss in every analysis (`docs/65`). Repair is now refused by default.
- **G-family.** The analysis ladder of paper 02, stored with `E` codes. The members this plan names:
  - **G1** (`E1`): binary drug indicator, `corCAR1`.
  - **G6** (`E1cr2`): G1 with CR2.
  - **G8** (`E2cr2`): D_{bc} at the true half-life, with CR2.
  - **G9** (`E7cr2`): D_{bc} at a half-life chosen by AIC on the interaction model, with CR2. Slightly liberal (about +0.005); the side analysis after the split.
  - **G10** (`E7null_cr2`): half-life chosen by AIC on the model without the interaction, then the interaction model refitted and tested with CR2. **The recommended analysis of paper 02** (Section 2.8). It was called "G9-null" in earlier drafts and the referee-variant runs (decision 5).
  - **Oracle benchmark.** An analysis given the true half-life (G3, G8). It is not available to a trialist. It serves only as the reference that prices the selection step (decision 4).
  - **G8 at a fixed half-life** (`E3fix_cr2`): a prespecified value of 0.25 or 1 week. **Dropped from paper 02** (decision 4).
- **E0.** Hendrickson's original analysis: `lmer`, random intercept, binary indicator, and the expectancy main effect where expectancy varies. It stays

in paper 02.

- **F-family.** The G3 formula plus η_t and $B_i^c \eta_t$, with CR2. Members:
  - **F1:** $t_{1/2}^a$ fixed at the 0.2-week policy value.
  - **F1A:** chosen by AIC with the interaction in the model.
  - **F1A0:** chosen by AIC without it.
  - **F1A0B:** F1A0 with a parametric-bootstrap calibration that repeats the selection.

  The family moves to paper 15.
- **Joint recommendation.** The single analysis advice that papers 02 and 15 give together (Section 4).
- **CR2.** The Bell-McCaffrey bias-reduced cluster-robust standard error, used with Satterthwaite degrees of freedom.
- **Scenario family.** A set of simulation cells sharing one departure from paper 02's original grid (Section 2.3).
- **Block (N, A, T, S).** The four blocks of `36-run-referee-variants.R`:
  - **N:** null cells.
  - **A:** alternative cells.
  - **T:** trend pathway.
  - **S:** patient slope variance.
- **Patient slope variance.** A patient-specific linear time trend unrelated to the biomarker (`sd_slope`). It is a heterogeneity stress, not a prognostic one, and stays in paper 02.
- **Two-MCSE band.** The interval of 0.05 plus or minus two MCSE: $[0.036, 0.064]$ at 1,000 replicates.
- **Hybrid, OL+BDC, CO.** The designs: open-label titration then blinded discontinuation then a brief crossover; open-label titration then blinded discontinuation; two-period crossover.

1. Decision

Paper 02 is narrowed to one question: how to represent carryover and choose the analysis half-life for the biomarker-treatment interaction when the biomarker is predictive only. The data come from the refined process under covariance moderation, meaning the biomarker-response correlation is the interaction mechanism.

Everything about a biomarker that is also prognostic moves to a new companion paper, provisionally paper 15. So does the F-family (F1, F1A, F1A0, F1A0B), whose expectancy terms exist to protect against a prognostic biomarker.

Recommended analysis (decided 2026-10-30). Paper 02 recommends **G10**:

- D_{bc}, with the analysis half-life chosen by AIC over $\{0.25, 0.5, 1, 2\}$ weeks on the model *without* the interaction

- the interaction model refitted by REML at that half-life
- the test of `bm:Dbc` by CR2 with Satterthwaite df

G9, which selects on the model *with* the interaction, becomes the comparator that shows what that choice costs. The reason is size. In the paired block N evidence, G9 is liberal against the oracle in every cell, while G10 cannot be distinguished from it (Section 2.8). G10 gives up 1 to 3 points of power for that. Among analyses that do not need the true half-life, it is the most powerful one that holds nominal size. That is the sense in which it is "the best model" in paper 02. It is not the most powerful analysis, and the paper must say so. Section 2.8 lists what has to be run before the paper can carry this claim.

A note on wording: the narrowed paper 02 does not use the published process at 58b32a9. Step coupling in that process removes the signal at any positive half-life (`docs/55`). Describe the scope as "the refined process under covariance moderation, with a predictive-only biomarker".

Rationale.

- The split removes two of paper 02's submission blockers outright:
  - the repaired strength-0.6 rows
  - the untested `bmc:t` remedy

  Both become properly scoped results in paper 15, where the 0.6 cells are the repair-artifact exhibit and the `bmc:t` check becomes a tested analysis.
- It cuts about 15 to 20% of the manuscript.
- It gives the prognostic material a coherent paper of its own. That paper is largely drafted already across `docs/47`, `docs/49`, `docs/56`, `docs/64`, `docs/65` and `docs/series/ch19`.

1a. Decisions needed before work starts

Five points emerged while mapping the split. Decisions 1, 2, 4 and 5 are resolved. Decision 3 still needs author confirmation; the detail is in the section cited.

1. **Which F-family member carries the joint recommendation (Section 4). Resolved 2026-10-30.** With G10 as paper 02's analysis, the expectancy-term counterpart is F1A0. That is G10's selection rule plus η_t and $B_i^c\eta_t$, in its bootstrap-calibrated form F1A0B. This agrees with CLAUDE.md's provisional recommendation.

   F1, which fixes the half-life at 0.2 week, is not the recommendation. In paper 02's cells it loses up to 10 points of power to a misspecified half-life.

2. **Large-N size evidence (Section 2.6). Resolved 2026-10-30: option (a).** Family 6, the $N \in \{140, 280\}$ cells, is entirely prog-PB. After the split, paper 02 has no size evidence above $N = 70$. The paper states the $N \leq 70$ limit, and no large-N run is made.

3. **Overlap between paper 01 Study 3 and paper 15 (Section 5.2).** Paper 01 already contains a prognostic stress test and a joint feasibility table (`tab:prognostic`). The recommendation is that paper 15 cite paper 01 for the DGP-level facts and present only the analysis-level consequences. Paper 01, which is near submission at grade B, would not be reopened.

4. **Role of the models that fix the analysis half-life. Resolved 2026-10-30.** Models that assume a known half-life lack face validity as candidate analyses. The three groups are handled as follows:

   - **(i) Oracle analyses at the true half-life (G3, G8). Kept**, labeled "oracle benchmark, not a candidate analysis" throughout. G8 is the reference that prices the selection step. The size evidence of Section 2.8 is stated against it.
   - **(ii) Analyses at a prespecified value (G8 at 0.25 or 1 week, `E3fix_cr2`). Dropped** from paper 02, together with every result and sentence about them. F1 leaves with the F-family. The cut-list consequences are in Section 2.2.
   - **(iii) Binary-indicator analyses (E0, G1, G6). Kept.** These fix the half-life at 0 implicitly. They stay as the published and default reference analyses.

   Cost of dropping (ii). Paper 02 loses its evidence that a fixed 1-week value stays within 2 points of the oracle at true half-lives of 0.5 to 2 weeks. A referee may still ask why the analyst should not simply fix the half-life. The answer the paper keeps is the binary indicator, which fixes it at 0. In family 1 it reaches 0.563 power against the oracle's 0.640 (G1 against G8). Prespecified nonzero values are not evaluated, and the Limitations should say so in one sentence.

5. **Name for G9-null. Resolved 2026-10-30: renamed G10.**

   - The stored code stays `E7null_cr2`. The display mapping in the Rmd (`m_lab`, `rv_lab`) and the ladder script map it to G10, as they already map `E` codes to `G` codes.
   - G9 is described as "G10 with the half-life chosen on the interaction model".
   - Record G10 in NOTATION.md and in paper 02's glossary.
   - The name "G9-null" survives only in stored data labels and in the referee-variant text being replaced.

2. Paper 02 after the split

2.1 Scope statement (Introduction and Methods)

Add one paragraph to the Introduction. Suggested text:

> The biomarker in this study is predictive only: it changes the drug response and nothing else. A biomarker that is also prognostic raises a different question, namely whether the interaction test confuses an association with the natural history or the placebo response with moderation of the drug effect. A constant prognostic shift is absorbed exactly by the biomarker main effect of every analysis here and cannot move the interaction estimate. Prognostic associations that vary over time, and the design-expectancy terms that protect against them, are the subject of a companion paper [paper 15].

Keep the absorption check for the constant shift (b_{TV}, "prog-TV-grow" in the stored data). It takes one sentence and answers the first question a referee will ask.

2.2 Section-by-section cut list

Status values:

- **Delete:** remove the section.
- **Trim:** remove the prognostic or F1 content only.
- **Rewrite:** recast the text without it.
- **Keep:** leave as it is.
- **Move:** relocate the material to paper 15.

Section (label)	Lines	Action
Setup chunk: `s28`, `s30`, `pdc`, `ceil` loads	122-129	Delete. These are the prognostic stress, random-effects, PD-check and ceiling data.
Setup chunk: `m_lab`, `focus`, `focus_cols`, `rv_lab`, `rv_method`	140-164	Trim. Remove F1, F1A, F1A0 and F1A0B. Remove `E3fix_cr2` ('G8-fixed'). Map `E7null_cr2` to 'G10' (decision 5).
Setup chunk: `rvT`	168	Delete. Keep `rvS`; see the slope row below.

Section (label)	Lines	Action
Abstract	230-273	Rewrite: (a) Methods: drop "or add expectancy terms (F1, F1A0B)". (b) Results: drop the F1 sentence, the trend-bias sentence and the strength-0.6 artifact sentence. (c) "Eight scenario families" becomes six; see 2.3. (d) Conclusions: recommend G10, not G9, and add "for a biomarker that is predictive only". (e) Methods: rung 4 is G10; G9 is the side analysis that selects on the interaction model. (f) Results: report the G10 size against the oracle, and G9's half-point excess as the cost of selecting on the tested model. (g) Methods and Conclusions: drop "fix it" from the side analyses and the sentence "A fixed 1-week analysis half-life with CR2 holds nominal size for at most two points of power" (decision 4).
Introduction, ladder overview	385-400	Trim. Remove the F-family sentence and the prognostic, trend and large-N items from the list of stresses. Add the scope paragraph from 2.1.

Section (label)	Lines	Action
Aims (`sec-aims`)	457-480	Trim. Remove "prognostic pathways, large samples, a biomarker that predicts the natural-history trend" from the second aim. Keep "patient-specific slope variance".
Table `tab-families`	482-500	Rewrite. Delete rows 5 (prognostic) and 6 (large N; every cell there is prog-PB). Row 7 becomes slope variance only. Renumber.
`sec-ladder-def`, table `tab-ladder-def`	643-672	Rewrite. Remove the pointers to the F-family. Redefine rung 4 as G10 (half-life chosen by AIC without the interaction). Label G3 and G8 "oracle benchmark, not a candidate analysis" (decision 4). Rename G9-null to G10 (decision 5).
`sec-g4` (rung 4)	769-802	Rewrite. Lead with null-model selection and why it keeps the test honest. Then present selection on the interaction model (G4, G9) as the variant whose size cost is measured. The existing F1A/F1A0 explanation (793-794) carries over.

Section (label)	Lines	Action
`sec-e0`	673-688	Keep. E0 contains the expectancy regressor `De` because Hendrickson's analysis does. That is part of the original analysis, not an F-family feature. Add one clause noting `De` is constant, and so dropped, under CO.
`sec-comparators`	803-839	Rewrite. Delete the F-family paragraph (807-831). G10 moves out of the side analyses into rung 4. Delete the "G8 at a fixed half-life" paragraph (decision 4). The only side analysis left is G9 (selection on the interaction model). Retitle "Side analysis: selecting on the interaction model".
`sec-pathways`	841-866	Delete, except one sentence on constant-shift absorption, which goes in the scope paragraph.
`sec-choice`	868-886	Keep. The earlier `bm:L` and `bm:tsd` models are a carryover question.
`sec-grid`	924-960	Trim. Remove the prognostic-cell clauses (924, 929, 950-955).
`sec-corrections`	1008-1058	Trim. The PD-refusal correction stays as a statement about the driver. Its prognostic-scenario evidence moves to 15 (1041-1044).

Section (label)	Lines	Action
Ladder-at-a-glance (`sec-glance`) and Fig. 1-3 captions	1072-1274	Rewrite after the ladder script is rerun (2.4). Remove F1 and F1A0B from the captions (1127, 1159) and from the prose (1206-1207, 1271).
`sec-rung1`, E0 isolation	1282-1580	Keep. The expectancy mentions here describe E0's `De` term, which stays.
Rung 4 null variants (`stage3-variants-null`)	2077-2125	Rewrite. Remove the F1 column and the 'G8 at 0.25 wk' and 'G8 at 1 wk' columns. The table becomes G8 (oracle), G9 and G10 in the eight Hybrid null cells.
`sec-alternatives`	2172-2341	Rewrite. Delete the fixed-half-life alternative (decision 4). Delete the F-family evidence (2176-2183, 2279-2339, the 'G9>F1' paired step). Move the G9-versus-F1A0B comparison to 15. What remains (null-model selection) has moved into rung 4, so fold any surviving text into `sec-g4` and delete the section.
`sec-scen-grids`	2353-2457	Trim. Remove the F1 columns (2381-2394, 2436-2448).

Section (label)	Lines	Action
`sec-dropout`	2458-2585	Trim. Remove the F1 columns (2472-2483, 2499-2500, 2514, 2524-2531, 2565-2566). Line 2533 cites the no-dropout reference of the prognostic stress job. Re-anchor it on a non-prognostic reference or drop the comparison.
`sec-prognostic` (all subsections, including `sec-feasibility`, `sec-feasible-grid` and "Larger samples")	2586-2939	Move to 15.
`sec-trend`: trend pathway	2940-3012	Move to 15 (table `stage5-trend`, Fig. 4).
`sec-trend`: patient slope variance	3014-3045	Keep as its own short subsection, "Patient slope variance", with G8 and G9 only. It is a heterogeneity stress, not a prognostic one: the slope is unrelated to the biomarker. Delete the fig4 trend contrast (3028-3040).
`sec-random`	3046-3090	Move to 15. It was fitted on the prognostic stress cells, 8 of 12 of them the repaired strength-0.6 cells, and was motivated by the artifact.
`sec-calibration`	3091-3178	Trim. Remove the F1 κ values (3142-3143).

Section (label)	Lines	Action
`sec-which`, ladder score narrative	3179-3262	Rewrite from the regenerated table B, with G10 as the reference analysis ("percent of G10 power"). "G9 has the largest cell size … under the trend pathway" no longer applies. Remove the F1 power-loss sentence (3250). "For a Hybrid trial G9 is the analysis to use" becomes G10, stated together with its power cost against G9.
Discussion, `sec-bought`	3265-3364	Trim. Remove the F-family and prognostic summaries (3331-3359).
`sec-recommend`	3365-3414	Rewrite. Recommend G10. Delete the fixed-1-week-half-life sentences (decision 4). Give the trade in one sentence: half a point of size bought back for 1 to 3 points of power. State the scope condition (predictive-only biomarker). Insert the joint-recommendation sentence (Section 4). Remove the trend caveat (3404-3406); it moves to 15.
"What the expectancy terms are for"	3415-3442	Move to 15. In 02, replace it with the one-sentence expectancy note in 2.5.

Section (label)	Lines	Action
Limitations	3443-3497	Trim. Remove the repaired-cell, prognostic-grid and large-N F-family sentences (3469-3488). Add the large-N gap (2.6). Add one sentence: analyses at a prespecified nonzero half-life were not evaluated (decision 4).
Conclusions	3534-3595	Rewrite items 4, 5, 6, 7 and 8. Item 4 becomes "choose the half-life on the model without the interaction". Item 7 recommends G10. Item 5 keeps only the interaction-model-selection finding (G9's size excess); the fixed-half-life finding is deleted (decision 4). Item 6 (feasibility) moves to 15. Item 8 loses the prog-PB and trend sentences and gains the scope condition.
Glossary	3596-3715	Trim. Remove F1, F1A, F1A0, F1A0B, the prognostic labels, the feasible-grid note and the half-life policy value (3598, 3625, 3642). Replace the "G9-null; G8 at a fixed half-life" entry with a G10 entry and an "oracle benchmark" entry.

Section (label)	Lines	Action
Reproducibility	3751-3878	Trim. Remove the jobs 28 and 30, scripts 30 and 31, and block T entries (3786-3792, 3829).

2.3 Scenario families after the split

New	Old	Content
1	1	Original 30-cell grid
2	2	Misspecified half-life and decay shape
3	3	Serial correlation
4	4	Patient-level dropout
5	7 (block S only)	Patient slope variance
S	S (block N: $c_{bm,PB} = 0$ cells only; block A)	Independent-seed Hybrid run

Block N of the referee variants holds four null cells at $c_{bm,PB} = 0.3$. They leave family S. The pooled G9 null figure in the abstract ("7,987 null fits … 0.055") must be recomputed without them. Check whether `m2-pooled-size.csv` (script 34) already pools only $c_{bm,PB} = 0$; the ladder script sets `pooled_null = c_pb == 0` for block N, but script 34 has not been checked.

2.4 Ladder script (`37-ladder-tables-figures.R`)

The script reads stored data only, so regenerating it is cheap. The changes are:

1. Drop family 5 (job 28 and the gap-fill sub `C1-ladder`), family 6 (sub `C2-largeN`), block T, and the block N cells with $c_{bm,PB} > 0$.
2. Drop F1 and F1A0B from `analysis_cols`, the paired steps (the 'G9>F1' row) and `side_x`. Drop them from the Fig. 2 and Fig. 3 analysis lists. Drop 'G8-0.25' and 'G8-1' the same way (decision 4), and relabel 'G9-null' as 'G10' (decision 5).
3. Make G10 rung 4 and the reference analysis:
   - Change `mean_power_pct_of_g9` to percent of G10.
   - Set the paired steps to G8 to G10 (rung 3 to 4) and G10 to G9 (side).
   - Move G9 to `side_x`.

   This needs the backfill of Section 2.8 first. Until then G10 exists in family S only.

4. Delete Fig. 4 (trend). Paper 15 will need its own version.
5. Rerun the script and check that tables A, B and C and Fig. 1-3 still satisfy the Rmd's `stopifnot` guards.
6. Re-read every inline `lv()`, `inc()` and `bv()` call against the new tables.

Rather than deleting code, add a `--scope predictive` switch that filters these out. Paper 15 can then run the same script with `--scope all` for its G-versus-F comparisons.

2.5 Expectancy note for 02

A referee will ask why the G-family omits a design covariate that varies in Hybrid and OL+BDC. Answer in one sentence in `sec-recommend`:

> With a predictive-only biomarker the G-family holds nominal size without the design-expectancy terms in every family examined; the terms matter when the biomarker may also be prognostic, which the companion paper [15] examines.

Do not claim more than that. The claim that F1 matched G9's power at every alternative belongs to paper 15, which carries the evidence.

2.6 Evidence gap the split opens

Family 6 (large N) consists entirely of prog-PB cells. After the split, paper 02 has no size evidence above $N = 70$ in the predictive-only case.

Decision (2026-10-30): state the limit; no large-N run. Paper 02 reports size at $N \in \{35, 70\}$ only. It makes no claim about larger trials. The text changes are:

- **Limitations.** Add: "Size was evaluated at $N = 35$ and $N = 70$. Behavior at larger samples was not examined for a predictive-only biomarker." This replaces the present partial statement ("measured only up to $N = 70$ with complete data") and the $N = 280$ sentences that move to paper 15.
- **Conclusions and `sec-recommend`.** Qualify the recommendation as holding "at $N \leq 70$". Most aggregated N-of-1 trials fall in this range.
- **Do not argue that size problems shrink at larger N.** The stored data do not support it. Mean model-based size moves from 0.033 to 0.028 (G1) and from 0.032 to 0.027 (G3) between $N = 35$ and $N = 70$, consistent with conservatism that comes from model misfit rather than small samples. CR2's small-sample refinements would fade at larger N, but the need for a robust standard error would not. One sentence in the Discussion can say this, without a claim about where the boundary lies.

The run considered and declined was a Hybrid null at $N \in \{140, 280\}$ with G8, G9 and G10, and model-based and CR0 arms of G10. Estimated at 1 to 3 hours on zeus. It remains available if a referee asks.

2.7 Effort for 02

- 1 to 2 days of rewriting.
- About 1 hour for the ladder script and checks.
- The G10 backfill (2.8):
  - half a day for the driver
  - an estimated 2 to 5 hours on zeus (untimed)
- No large-N run (decision 2).

2.8 G10 as the recommended analysis: evidence and backfill

Evidence in hand. Block N, job 34: Hybrid, 5,000 replicates per null cell, with G8, G9 and G10 fitted to the same trials. The table shows the four cells with $c_{bm,PB} = 0$, the ones paper 02 keeps.

Cell	G8 (oracle)	G9	G10	G9 − G8 (paired SE)	G10 − G8 (paired SE)
$t_{1/2} = 0.5$, $N = 35$	0.048	0.052	0.047	+0.004 (0.0016)	\$-\$0.001 (0.0015)
$t_{1/2} = 0.5$, $N = 70$	0.051	0.054	0.051	+0.002 (0.0014)	\$-\$0.000 (0.0014)
$t_{1/2} = 1$, $N = 35$	0.051	0.060	0.051	+0.009 (0.0019)	\$-\$0.000 (0.0017)
$t_{1/2} = 1$, $N = 70$	0.046	0.051	0.045	+0.004 (0.0017)	\$-\$0.001 (0.0016)

The two-MCSE band at 5,000 replicates is $[0.044, 0.056]$. The results are:

- **G9 is liberal against the oracle in every cell.** It is in all 8 block N cells, including the prognostic ones, and 7 of those 8 are above 2 paired SE.
- **G10 is within 1 SE of the oracle everywhere.**
- **The standard errors agree with this.** κ is 0.89 to 0.97 for G9 and 0.94 to 0.99 for G10.
- **Power.** G10's cost is 1 to 3 points, from block A: 8 alternative cells at 1,000 replicates, Hybrid only. In ladder table B it reaches 96.6% of G9's power.

What is missing. G10 was fitted only in family S (blocks N and A, Hybrid, $N \leq 70$). A recommended analysis must be shown in every family the paper reports:

- family 1, the 30-cell grid over CO, Hybrid and OL+BDC (job 27)
- family 2, misspecified half-life and decay shape (job 21 cells and gap-fill B1, B2)
- family 3, serial correlation (gap-fill B4)
- family 4, patient-level dropout (gap-fill A1, plus the A2 size diagnostic)
- the slope-variance cells (block S)

Backfill run. Write a new driver, `40-run-g9-null-backfill.R`:

- It loops over the cell grids of `27-run-tier1-g-vs-f-e0.R`, `32-run-g9-gap-fill.R` (sub-blocks A1, A2, B1, B2, B4), `06-g-vs-f-comparison.R` (the four Stage 4 cells) and block S of `36-run-referee-variants.R`.
- For each (cell, replicate) it fits `E7null_cr2` and `E7cr2`. `fit_spec_s7()` already implements both.
- **Pairing.** Each driver seeds a replicate as `set.seed(base + r)` before generating data, so rerunning on zeus with the same seeds regenerates the same trials. G9 is refitted only as a check: its estimate must match the stored value to 10^{-8} in every replicate.
- **If the check fails** (a DGP or image change since the original job), the stored G9 cannot be paired. Rerun G8, G9 and G10 together in that sub-block and pair within the new run.
- **Size:** roughly 55,000 replicates.
- **Cost:** about two G9-sized fits per replicate. Estimated at 2 to 5 hours on zeus's 8 cores, but untimed; run `--time` first.

What would overturn the recommendation. Two outcomes would:

- a family in which G10 leaves the size band while G9 does not
- a family in which G10 loses much more than 3 points of power

The CO design deserves attention. There the candidate half-lives differ least in their fixed-effect fit, so null-model AIC carries little information about the half-life.

Expected grade after the cut: B, with the front matter (cover letter, title page, declarations) still outstanding as in `docs/66`.

3. Paper 15 outline: prognostic biomarkers and the interaction test

Working title: *When the biomarker is also prognostic: bias in the biomarker-treatment interaction test of aggregated N-of-1 trials and two remedies.* Direc-

tory: `analysis/report/15-prognostic-biomarker/`. Target: the same venue as 01 and 02, as a third companion, or a methods journal on its own.

Section	Content	Source (existing)
1. Introduction	Predictive versus prognostic biomarkers; why the interaction test can confuse the two in N-of-1 designs with varying expectancy	`docs/49`, `ch19`, lexicon `docs/29`
2. Taxonomy of pathways	Constant shift (absorbed); correlation pathways prog-BL, prog-TV-const and prog-PB; mean-shift trend pathway (unbounded)	02 `sec-pathways`; `tidyverse/R/functions.R` trend block
3. Feasibility	Closed-form joint bound on the prognostic correlation and c_{bm} (Schur complement); feasible grid	`docs/65`; scripts 30 and 31; 02 `sec-feasibility`. **Paper 01 already has a joint-ceiling table** (`tab:prognostic`, `18-prognostic-joint-ceiling.R`). Cite 01 for it rather than duplicating; see 5.2.
4. The repair artifact	Strength-0.6 cells: power loss from silent PD repair, not confounding	`docs/65`; jobs 28 and 30; 02 `sec-prognostic` "strength-0.6" subsection; `sec-random`
5. Analyses	G-family (G8, G9) against the F-family (F1, F1A, F1A0, F1A0B); expectancy terms η_t, $B_i^c\eta_t$ (regressors `De`, `bmc:De`); why they absorb the expectancy-scaled on/off contrast under prog-PB	`docs/56`, `docs/57`, `docs/62`, `docs/64`; 02 `sec-comparators` and "What the expectancy terms are for"
6. Correlation pathways: results	Size and power at strength 0.3 and on the feasible grid; large N under prog-PB	Job 28; gap-fill `C1-ladder` and `C2-largeN`; 02 tables `stage5-*`

Section	Content	Source (existing)
7. Trend pathway: results and the `bmc:t` remedy	Bias of every analysis; linear and quadratic `bmc:t`; cost when there is no trend	Block T (job 36); **new run** (3.1)
8. G-family against F-family without a prognostic effect	The F-family's cost when unneeded; F1's fixed-half-life penalty versus F1A0B	Job 27 (`tier1-g-vs-f-e0`), job 21 (`g-vs-f-comparison`); 02 `sec-alternatives` F-material
9. Recommendation	Joint recommendation (Section 4)	
Appendix	Constant-shift absorption proof; repair mechanics	`docs/65`

3.1 New simulation for paper 15

`bmc:t` block.

- Add specifications `E2cr2_bmt`, `E7cr2_bmt` and `F1_bmt` to `simulation-core.R`: the existing formula plus `bmc:t`. For G9, add the term to both the AIC-selection formula and the refit formula. Optionally add quadratic variants.
- Add a new block to `36-run-referee-variants.R`. Cross the trend pathway $b_{TV}^{\text{tr}} \in \{0, 0.3\}$ with $c_{bm} \in \{0, 0.45\}$ at Hybrid, $N = 70$, $t_{1/2} = 0.5$, 1,000 replicates, plus a 5,000-replicate null trend cell for size.
- Estimate about 30 to 45 minutes on zeus. Block T took 8 minutes for 2 cells and 3 specifications.

A local pilot on 2026-10-29 (scratch script, 300 replicates, G8 only) gave the following:

- The null bias of +0.26 fell to +0.10 with linear `bmc:t` and to +0.00 with quadratic.
- Power fell from 0.96 to 0.94 and 0.92.

The DGP's trend follows the TV Gompertz course, not a straight line, so linear `bmc:t` is expected to be only a partial remedy.

3.2 Effort for paper 15

About 1 to 2 weeks:

- Drafting is mostly assembly from the sources above.
- One new simulation block (about 1 hour of compute plus half a day of code).
- The ladder script runs with `--scope all` for the G-versus-F figures.

4. Joint recommendation (both papers, CLAUDE.md, NOTATION.md)

The two papers must give one answer between them. Present the F-family as the G-family plus η_t and $B_i^c \eta_t$ (regressors `De` and `bmc:De`):

- **Paper 02** settles the carryover layer with G10. That means an exposure-weighted predictor, with the half-life chosen by AIC on the model without the interaction, and CR2 with Satterthwaite df.
- **Paper 15** adds the expectancy layer when the biomarker may be prognostic, and a `bmc:t` layer if the biomarker may predict the natural-history trend.

Resolved 2026-10-30: the F-family member is F1A0, and its calibrated form is F1A0B. F1A0 is G10's selection rule with η_t and $B_i^c \eta_t$ added. Choosing it makes the two layers nest exactly: G10 and F1A0 differ only in the expectancy terms. It also matches CLAUDE.md's provisional F1A0B.

Two things remain for paper 15 to show:

- whether F1A0B's bootstrap calibration is needed once the selection excludes the interaction (`docs/64` found no Type I inflation)
- what F1A0 costs against G10 when the biomarker is not prognostic

F1 at the 0.2-week policy value is not the recommendation. It loses up to 10 points of power where that value is wrong (family 1).

Recommended wording:

> For a biomarker believed to be predictive only, use G10: the exposure-weighted predictor with the half-life chosen by AIC on the model without the interaction, tested with CR2 (paper 02). Where the biomarker may also be prognostic, add the design-expectancy terms η_t and $B_i^c \eta_t$ (F1A0, paper 15). Where it may predict the rate of natural-history change, also add a biomarker-by-time term.

Once decided, update:

- the CLAUDE.md "Recommended analysis for the interaction (F1)" paragraph
- NOTATION.md (analysis IDs; the new paper number)
- `docs/66` (portfolio table)
- `docs/60` Part 2 (paper 02 plan)

5. Cross-paper effects

5.1 Paper 01

Paper 01 cites paper 02. Check that none of its citations points at material that moves. Grep for "companion" in 01. As of 8e810ca9d, 01's own prognostic content is its Study 3 stress test, which is independent of 02.

5.2 Overlap between paper 01 Study 3 and paper 15

Paper 01 already contains:

- a stress test under a prognostic biomarker (Study 3, `tab:prognostic`)
- a joint feasibility table (`18-prognostic-joint-ceiling.R`)
- the statement that the stop-on-non-PD default came from "a prognostic scenario" with correlation 0.6 (lines 612-618)

Paper 15 must cite 01 for the DGP-level facts: the bound exists, and repair is refused. It should present only the analysis-level consequences. Paper 01 is at grade B and should not be reopened for this.

5.3 Portfolio count

The split adds a fifteenth manuscript. `docs/66` recommends consolidating toward about nine. The split is justified because the boundary is scientific, not editorial. It should be recorded alongside the merges already proposed there: 13 with 14, 08 into 10, 12 into 01, and 11 into 01 or 03.

5.4 Deferred: bootstrap-calibrated selection (G9-WB), a paper 10 pilot

Status: deferred (2026-10-30). This is not part of the paper 02 work. Paper 02 proceeds with G10 whatever the outcome.

Question. Can one calibration step fix both size errors that rungs 2 and 4 fix separately?

- the conservatism of the model-based standard error (CR2's job)
- the liberal excess from choosing the half-life on the interaction model (null-model selection's job)

If it can, G9's selection rule, which is 1 to 3 points more powerful than G10, could be used at nominal size.

Why a parametric bootstrap will not do. Resampling from the fitted working model (random intercept plus `corCAR1`) reproduces the working model's error structure. That model is misspecified relative to the DGP, which is the source of the conservatism, and under it the model-based SE is correct by construction. Such a bootstrap would therefore fix only the selection excess, which G10 already fixes more cheaply.

Candidate: G9-WB. A null-imposed wild cluster bootstrap of the whole procedure (Cameron, Gelbach and Miller 2008). Each resample:

1. Fits the model without the interaction.
2. Multiplies each patient's residual vector by one random weight. Use Webb six-point weights, given $N = 35$ clusters.
3. Rebuilds the outcome.
4. Repeats G9's AIC half-life selection on the interaction model.
5. Refits and records the model-based t for `bm:Dbc`.

The observed t is compared with the resampled distribution. Patient-level resampling keeps the true within-patient covariance, so no sandwich estimator is needed.

Pilot design.

- **Cells:** family S. That is block N (8 null cells, 5,000 replicates) and block A (8 alternative cells, 1,000 replicates): Hybrid, $N \in \{35, 70\}$.
- **Arms:** G9-WB alongside G8, G9 and G10, on the same trials.
- **Bootstrap:** the warp-speed method (Giacomini, Politis and White 2013). Each Monte Carlo replicate draws one bootstrap resample, and the critical value comes from the pooled resamples of the cell, as F1A0B does. This validates the procedure's size and power on average, not single-trial p-values.
- **Driver:** a new specification in `simulation-core.R` and a new block of `36-run-referee-variants.R`.
- **Cost:** half a day of code. About 1 to 3 hours on zeus, at roughly twice the cost of G9 per replicate (estimate, untimed).

Adoption rule (exploratory, stated before the run). Adopt G9-WB only if both of these hold:

- its null rejection lies in the two-MCSE band in all 8 block N cells
- its mean power exceeds G10's by at least 2 points over block A

Otherwise record the negative result in paper 10 and stop.

Reasons for low expectations.

- F1A0B, the nearest existing bootstrap-calibrated analysis, reached 92.9% of G9's power in paper 02's ladder, with a largest cell size of 0.070.
- `docs/64` found no power gain for F1A0B over F1.
- The most G9-WB could win is the 1 to 3 points that separate G9 from G10.

Home. Paper 10 (interaction-test calibration). Its current argument is mean-model misfit with CR2 as the remedy, and G9-WB is a competing remedy. A trialist would run the full bootstrap (about 999 resamples times 5 fits, a few minutes per trial), so a positive result would also need a single-trial calibration check that the warp-speed pilot cannot give.

6. Order of work

1. Confirm decision 3. Decisions 1, 2, 4 and 5 are settled.
2. Write and time `40-run-g9-null-backfill.R` (2.8) and run it on zeus. Check the G9 pairing guard before using any result.
3. Confirm that G10 holds the size band in every family. If a family overturns the recommendation (2.8), stop and revisit before rewriting.
4. Add `--scope` and the G10 reference to the ladder script, then regenerate the predictive-only tables and figures.
5. Cut and rewrite paper 02 per 2.2. Re-verify every inline number. Run `perl tools/notation-lint.pl`. Render through `tools/render.sh`.
6. Create `analysis/report/15-prognostic-biomarker/` and move the excised Rmd sections into a first draft, keeping their chunks working against the same data.
7. Implement and run the `bmc:t` block (3.1). Include F1A0 alongside G10.
8. Update CLAUDE.md (the recommended-analysis paragraph: G10, with F1A0/F1A0B as the prognostic-safe form), NOTATION.md, `docs/60` and `docs/66`.

7. Unverified assumptions in this plan

- The line ranges come from a grep of section headers and F1 and prognostic mentions, not a line-by-line read. Some cross-references outside the listed ranges will also need cutting.

- I have not checked whether `m2-pooled-size.csv` excludes the $c_{bm,PB} = 0.3$ null cells (2.3).
- The conservatism-from-misfit reading in 2.6 rests on two sample sizes ($N = 35$, 70), mostly at 500 replicates per cell. It is consistent with the data but not a demonstrated trend.
- The cost of the `bmc:t` block is scaled from block T's runtime. Its findings rest on a 300-replicate G8-only local pilot.
- G10's standing as the best-sized analysis rests on the 8 block N null cells and 8 block A alternative cells (Hybrid, $N \leq 70$). Its behavior in CO, OL+BDC, dropout, misspecified-shape and serial-correlation cells is unknown until the backfill runs.
- The backfill's pairing assumes that rerunning the stored seeds on zeus regenerates identical trials. Any DGP or image change since the original jobs breaks this. The G9 refit check (2.8) detects it; nothing has verified it yet.
- The backfill cost (2 to 5 hours) is an estimate, not a timing.
- The deferred G9-WB pilot (5.4) has never been tried in this project. Two things about it are unverified:
  - that a wild cluster bootstrap with Webb weights behaves well at $N = 35$ with eight visits per patient
  - its cost
- The deferred pilot is named "G9-WB", not G10, because G10 is now the recommended analysis (decision 5).